BATCH & BUILD · STEM LEARNING SHEET · AGES 8–12

Wind Power: Catching the Invisible (Mini Windmill kit)

THE SCIENCE

The sun warms the Earth unevenly. Land heats faster than water, and dark ground heats faster than snow. Warm air is lighter, so it rises, and cooler air slides in underneath to take its place. That sliding air is wind.

Anything that moves carries energy, called kinetic energy. When wind pushes on slanted windmill blades, part of that push turns sideways and makes the blades spin. Now the energy is in the spinning shaft.

A generator turns that spin into electricity. Inside are magnets and coils of copper wire. When a coil spins near a magnet, electricity flows through the wire. Real wind turbines do exactly this, and their blades can be longer than a school bus. In your kit, a tiny LED light shows the electricity. An LED only lets electricity through one way, so if it doesn't glow, try flipping its wires.

Wind is renewable, which means nature keeps refilling it, and turbines don't make smoke while they work. The catch is that wind doesn't blow all the time, so it's usually combined with other kinds of power.

KEY WORDS

Wind: air moving from one place to another.

Energy: the ability to make something move or change.

Kinetic energy: the energy of something that's moving.

Turbine: a machine with blades that spin when wind or water pushes them.

Generator: a device that turns spinning motion into electricity.

Renewable: a resource that nature keeps refilling, like wind or sunlight.

Circuit: a complete loop that electricity can flow around.

EXPERIMENT – WHICH BLADE COUNT WINS?

Question: Does adding blades make a windmill spin faster?

1. Build three blade sets for your windmill, with 2, 4, and 6 blades, all the same size and bend.
2. Put a sticker on one blade.
3. Set the fan to low and place it the same distance away each time.
4. Count how many times the sticker passes the top in 10 seconds. Do 3 tries per blade set and find the average.
5. Record your results in a table and circle the winner.

Think about it: Did more blades always mean faster? Real turbines usually use 3 blades. Why might that be a good balance?

SAFETY

An adult snips the skewer's sharp tip and does all the cutting. Keep fingers, hair, and faces away from spinning blades and fans, and use a hair dryer only on the cool setting. Never connect the project to a wall outlet. Keep small parts like LEDs and wires away from kids under 3.

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CHECK YOUR UNDERSTANDING

1. What makes wind happen?
2. What kind of energy does moving air have?
3. What two things inside a generator work together to make electricity?
4. Is wind a renewable or nonrenewable resource?
5. Why are a windmill's blades slanted instead of flat?
6. Name one problem with wind power.
7. True or false: an LED must be connected the right way around to light up.

ANSWER KEY (FOR GROWN-UPS)

1. The sun heating the Earth unevenly, so warm air rises and cool air moves in.
2. Kinetic energy.
3. Magnets and coils of wire.
4. Renewable.
5. So the wind's push turns into a sideways spin.
6. The wind doesn't always blow.
7. True.

Name: _____